

Six-slide brief

Use the **official six-slide template and its headings**. The PDF is the idea-round upload. Bold text below is slide copy. Label proposed capabilities **PLANNED**; label the existing prototype **PARTIAL**. Use no survey, invented prototype URL, unsourced cost percentage, or projected lives saved. Reference IDs point to DOC 3. [P0](#)

Slide 1 — Title

Layout: Follow the reference title fields. Put the idea title in the largest type and the child line below it.

Field	Exact text
Problem statement ID	SIH26037
Idea title	Safety-Checked Adaptive Path Planning for Unstructured Indian Roads
Theme	Smart Vehicles
Category	Software
Team ID	[TEAM ID]
Team name	[TEAM NAME]
One-line idea	The car watches and listens, guesses what might move, and drives only when it has room to stop.

Limit: Title ≤10 words (uses the PS's own keywords); one-line idea as written. Source: R01.

Slide 2 — Proposed Solution

Layout: Left solution bullets; central hub **D1**; right “It addresses the problem by”; narrow platform strip and tagline below. Mark D1 **PLANNED SYSTEM**.

Left column — copy verbatim; max 45 words

- **One car, one loop:** sense → understand → decide → act → prove.
- **Mixed traffic:** detect autos, pushcarts, pedestrians, animals and uncertain road edges.
- **Several futures:** predict short-term, non-lane-based movement.
- **Only checked moves:** brake and report **BLOCKED** when no safe move passes.

D1 central hub — max 38 label words: One idea contains these five layers and **13 named parts**. Group them into roughly ten visual bubbles without dropping a part:

Layer	Parts
SENSE	Sensors
UNDERSTAND	Perception ML; Hearing; Gesture reader; Tracker; Prediction ML
DECIDE	Planner; Safety checker; Safety monitor
ACT	Control; Signals & driver screen
PROVE	World; Test & learn loop

Right column — copy verbatim; max 43 words

- **No lane markings:** check road, shoulder and unknown space.
- **Informal merging:** consider yielding and non-yielding responses.
- **Sudden movement:** reserve space for crossing people and cattle.
- **Unexpected obstacles:** preserve a visible stopping path.
- **Closed loop:** the car’s action changes the next traffic scene.

Platform strip — max 19 words: MATLAB/Simulink ego stack · SUMO reactive traffic · CARLA 0.9.15 Indian scenes · real Indian video replay

Tagline — max 13 words: Learning ranks moves. An independent physical check alone permits movement.

Source: R01; [P13](#).

Slide 3 — Technical Approach

Layout: Top expert line; left tech-stack logo wall grouped by the five layers; **D2** centre; **D3** right; prototype strip at bottom. Use a clear **PLANNED** label over the joined pipeline.

Top line — exact: One Simulink ego stack combines Indian-road sensing and prediction with a learned manoeuvre scorer, an independent movement verifier, and measured closed-loop tests.

Tech-stack wall — max 52 words total

- **SENSE:** CARLA camera/LiDAR/radar; Automated Driving Toolbox.
- **UNDERSTAND:** Python/PyTorch → ONNX → Deep Learning Toolbox; trackerGNN.
- **DECIDE:** Navigation Toolbox; Stateflow; MATLAB Coder.
- **ACT:** Simulink control; Vehicle Dynamics Blockset.
- **PROVE:** [drivingScenario](#); official R2026a SUMO blocks; CARLA scene replay.

D2: One decision: **observe** → **track/predict** → **rank 8–16 moves** → **check swept space and stopping distance** → **drive or brake**. Show the safety monitor’s veto arrow. Max 24 words inside the diagram.

D3: Label “**Five-model target core; nine later models**”, never “14 models running.” Exact model labels:

- **Core target:** YOLOX object detector; DeepLab v3+ road/pothole mask; pose-based hand-signal reader; ego-conditioned top-K motion predictor with conformal coverage; move scorer.
- **Later tier:** lidar 3D detector; BEV free-space/occupancy; sign and Indian-text reader; siren/horn recogniser; visibility estimator; driver watcher; crossing-intent model; RL policy **comparison only**; learned traffic agents.

YOLOX and DeepLab exist **offline**; the other core models and the joined loop are **planned**. No AI model can move the car without the verifier. [P2](#), [P15](#)

Prototype strip — max 22 words: **PARTIAL today:** internal [sihDemo](#) runs S1 and sparse S3. **Public prototype link:** [INSERT VERIFIED URL AFTER UPLOAD].

Source: R01, R20–R28; [P2](#).

Slide 4 — Feasibility and Viability

Layout: Four boxes as in the reference: Financials, Feasibility, Viability, Challenges → Mitigation. Put **D5** as a compact bottom timeline. Use the **cost component of D7** in Financials.

Financials — max 29 words: **Software ₹0** (KIET MATLAB licence; open-source CARLA, SUMO). **Training ₹0** on KIET DGX A100. Backup cloud A100 ~\$0.43–1.09/GPU-hour (GetDeploying, Thunder Compute, Sep 2026).

D7 graphic: use D7’s cash-cost table (sourced) and its effort-split pie, labelled “**planned effort split**” — never as a cost pie.

Feasibility — max 32 words: **Two builders, ten-week plan.** W2 gates: 100 ms fast-tier planning and one custom prop plus vehicle spawning. W8 gate: five scenes and full camera-in-loop timing.

Viability — max 30 words: **Real starting point (PARTIAL):** MATLAB planner completes S1 (610 m); **348/348** automated tests; real-map pipeline; offline Indian road mask **96.0% drivable IoU**. Open failures are disclosed in Q&A (DOC 4), not on the slide.

Challenges → Mitigation — max 11 words per row

Challenge	Mitigation
Slow checking	Profile, reduce candidates, compile and retest full-loop timing.
CARLA assets	Start source build W1; prove prop and vehicle W2.
Render-to-real gap	Measure per-class recall on held-out renders and Indian footage.

D5: Ten weekly gates; distinguish A's brain/safety/evaluation track from S's world/assets/perception track. Add a red caption: “**Two detailed CARLA scenes replace specified RoadRunner scenes; no RoadRunner licence.**”

Source: [P2](#), [P15](#), R27–R28.

Slide 5 — Impact and Benefits

Layout: Stakeholders left; large sourced problem numbers centre; **D4** five-scenario strip right; **D6** coverage/beyond panel at bottom. Numbers describe the **problem**, not an impact achieved by this proposal.

Stakeholders — max 24 words

- **Road users:** safer decisions are the research goal.
- **ADAS and AV teams:** reproducible Indian-road stress tests.
- **Researchers and judges:** contact, progress, comfort and timing evidence.

Large numbers — exact labels

- **4,80,583 crashes · 1,72,890 deaths — India, 2023.** R03
- **2,161 pothole-related deaths — India, 2023.** R03
- **3,383 stray-cattle crashes — Haryana, preceding five years.** R04
- Small footnote: **50 lakh stray cattle — 2019 livestock census; not a road-encounter rate.** R05

D4: Exact PS scenes: **unmarked village road; unsignalled urban intersection; highway merge with slow vehicles; dense market; sudden cattle crossing.** Label **PLANNED CLOSED-LOOP TESTS.**

D6 — max 31 words: **90 difficulties catalogued → ten composable families → 25 priority probes.** Planned proof: **replanning latency, path smoothness, scenario completion rate**, plus contacts and false stalls. **RoadRunner substitution remains a deviation.**

Optional small behaviour note, if legible: “**Patna sample: 26% rolling crossings; site-specific, not a national rate.**” R06. Do not claim an estimated crash reduction.

Source: R01, R03–R06; [P1](#).

Slide 6 — Research and References

Layout: Left: short reference list; right: **D7 competitor component**; bottom: dataset ribbon, prototype placeholder and RoadRunner disclosure. Keep the official “Research and References” heading.

Visible references — max 65 words: **MoRTH 2023; Haryana Assembly cattle answer; IDD; IDD-PeD; METEOR; Kruse et al. 2022; Mosaic 2026; MathWorks SUMO/CARLA documentation.** Use small `[Rxx]` tags linked to DOC 3.

D7 competitor table: Use **✓ documented**, **P proposed**, **? unverified**. Do not use **x** to assert an undocumented absence.

Organisation	Indian traffic/scenario work	Independent movement checker	Paired Indian closed-loop metrics
Our system	✓ catalogue; P executable suite	P	P
The Syndicate*	?	?	?
ARAI	✓	?	?
WMG Safety Pool India	✓	?	?
Swaayatt Robots	✓ company-described work	?	?
Minus Zero	✓ company-described work	?	?

*Teammate-supplied description says stock CARLA, YOLOv11 + LiDAR, a three-tier planner and three scenes; **independent evidence was not found in the research pack.** Do not convert that description into ticks or crosses. ARAI and WMG already have Indian scenario work; do not claim “first Indian test suite.” [P1](#), [P3](#)

Dataset ribbon — max 22 words: **IDD family · DriveIndia · FGVD · TIAND · METEOR · Chennai SPT · RoadText · simulator data.** Access and licences are checked before training.

Bottom disclosure — exact: RoadRunner is unavailable: two planned detailed CARLA scenes substitute for, but do not literally meet, that PS clause.

Prototype link: [INSERT VERIFIED PUBLIC URL]; otherwise print “Internal prototype available on request”. Source: R01, R09–R19, R29–R32.

Keyword glossary — team use only

Term	Plain meaning
Adaptive path planning	Reconsidering the car’s path as the scene changes.
Unstructured Indian road conditions	Roads and traffic where lanes, edges and behaviour may be irregular.
Multi-sensor	Combining camera, LiDAR, radar and vehicle-state measurements.
LiDAR	Laser distance measurements that give obstacle shape and location.
Radar	Radio sensing that helps measure distance and closing speed.
Perception ML	Models that find road users, road surface and hazards.
Hearing	In this plan, simulated horn/siren direction and class events.
Gesture reader	Camera pose cues for raised hands and traffic direction.
Tracker	A continuing position and uncertainty record for each observed actor.
Short-term motion prediction	Possible actor positions over the next few seconds.
Ego-conditioned	Forecasting an actor given a proposed action by our car.
Non-lane-based movement	Sideways or crossing motion that does not follow a lane centre.
Conformal coverage	A calibrated method for sizing predicted occupied regions.
Learned manoeuvre scorer	A model that ranks a fixed set of candidate moves.
Independent movement verifier	A separate geometric and braking check that alone permits motion.
Safety monitor	A watchdog that cancels permission when data or control is faulty.
BLOCKED	An explicit state when no checked move exists.
Real-time replanning	Meeting a measured decision deadline while driving.
Closed-loop validation	The car acts, traffic responds, sensors update, and the next decision uses that update.
Replanning latency	Time from a scene snapshot to a checked replacement path.
Path smoothness	How sharply the driven path bends or acceleration changes.
Scenario completion rate	Share of runs that reach the goal under the declared rules.
ODD	The roads, speeds, weather and traffic conditions within which the system is tested.